BIOMEDICAL ROBOTICS · ASSIGNMENT 1

Steering a cursor with four muscles

Today, in the lab

One person per group wears the MCI and records a reaching session: about 15 min of setup and calibration, then 15 min of play. In the meanwhile, the others start to work on pre-recorded data.

Then, by yourself

You get your recording and a code skeleton the goal is to rebuild the cursor from the raw EMG and compare them.

Deadline: **03/11/2026.**

From your session to one .mat file

make_group_mat

Takes the session folder saved by the MCI software (Acquisizioni + Calibrazione) and packs everything you need into one file.

```
make_group_mat(sessionFolder, ...
'Cometa', 'Group_A.mat')
load('Group_A.mat')
```

What you find in the workspace

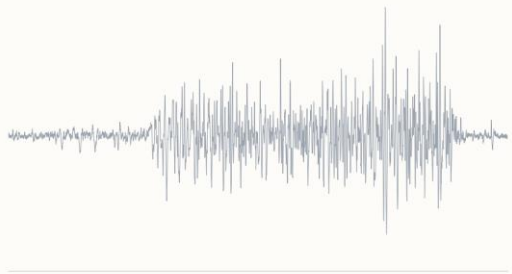
Variable	Contents
centerout	table, one row per sample: t, raw_R_biceps … raw_R_trapezius, env_R_biceps … env_R_trapezius, cursor_x_px, cursor_y_px, cursor_x_cm, cursor_y_cm
centerout_targets	table, one row per target: target_x_px, target_y_px, spawn_time, hit_time, reach_time, overshoots
random, random_targets	the same, for the random task
info	fs, muscles, screen size, cm_per_px, calib_max, gain_cm_s, W, map_x, map_y

env_ columns = the activation the MCI computed live (0–1)

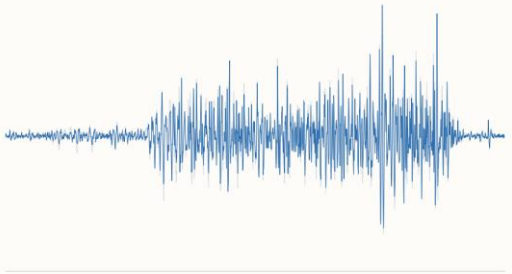
The assignment in three parts

Part 1 · The pipeline to obtain envelopes

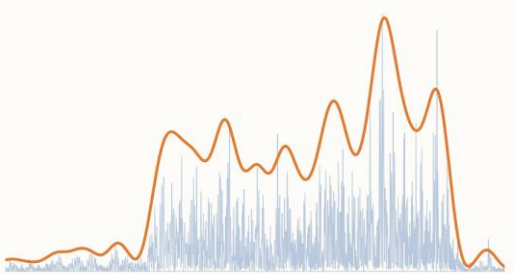
Raw EMG



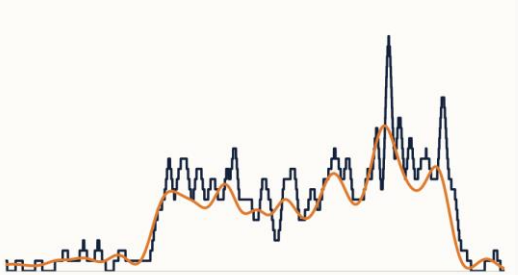
Band-pass 30–450 Hz



Rectify + low-pass 5 Hz



Yours (orange) vs MCI (black)



Part 2 · Fit the map

Use the gains and the envelope to create the map that goes from muscle intensity to cursor velocity.

Part 3 · Cursor

After applying the MCI map to get the cursor velocity, integrate it into a position, animate it with the targets.

Compare it with the cursor that was logged

Bonus: use your cursor to drive a simulated 2-link robot arm (Jacobian) .

The map, the gains and the numbers you need

The map is given

$$\text{velocity} = \text{gain} \cdot \text{env} \cdot W'$$

gain = info.gain_cm_s = 50 cm/s.
W = info.W: one row per axis, one column per muscle.

W	R biceps	L biceps	L trap	R trap
x (right)	+1	−1	0	0
y (down)	0	0	−1	+1

The same in words: info.map_x and info.map_y. Velocity is in cm/s: integrate it in cm, then convert.

Good to know

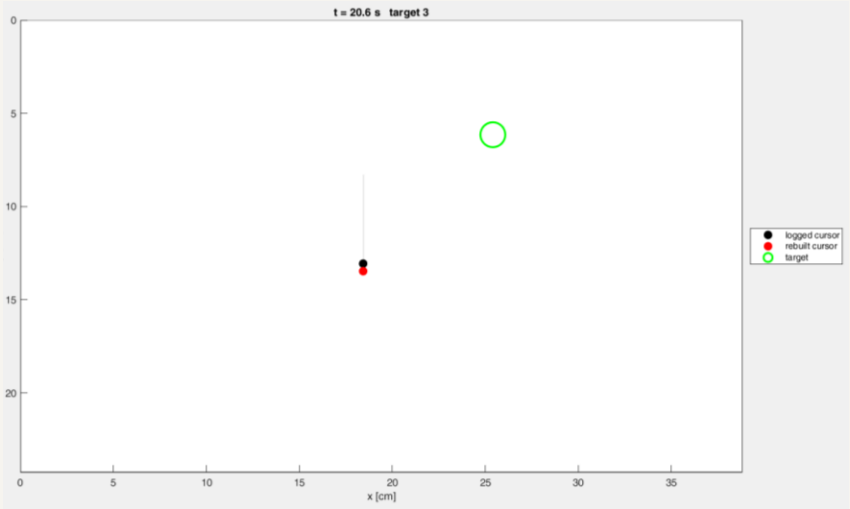
What	Value
Sampling	2000 Hz (info.fs)
Screen	2560 × 1600 px = 38.8 × 24.3 cm
px → cm	× info.cm_per_px (≈ 0.0152 cm/px)
Axes	origin top-left, x right, y down
Center target	(1280, 800) px
Raw EMG	already band-passed 10–500 Hz by the hardware
env_ columns	(envelope − rest) / max; the max can be raised live
Downsample	only after the envelope, never on raw EMG
Center-out	48 targets: 24 out, 24 back to the center
Random	100 s; the last target may never be hit (NaN)

What to hand in

One folder per group

```
Group_#_Assignment_1/
├── *.m            commented code
├── Group_#.mat    your data
└── Group_#.pdf    figures
```

Part 3 · Your cursor vs the logged one, with the target



Part 1 · Your envelope vs the one the MCI computed live (R biceps)

